
Measurement Theory and Operationalization

Advanced Topics: Methodology, Philosophy, and Research Design

1. Basic Concepts

Measurement: Assigning numbers or symbols according to rules (Stevens 1946).

Construct: Theoretical concept, not directly observable (e.g., intelligence, satisfaction).

Operationalization: Translating a construct into measurable indicators.

2. Scale Levels (Stevens)

Nominal: Equal/not equal. Categories without rank. Mode only.

Ordinal: Rank order, but unequal intervals. Median allowed, mean problematic.

Interval: Equal intervals, no absolute zero. Mean and SD meaningful.

Ratio: Equal intervals + natural zero. All operations allowed.

3. Reliability

Reliability = measurement precision / consistency.

Test-retest, parallel-test, split-half, internal consistency (Cronbach's alpha > 0.7 acceptable).

Interrater reliability: Cohen's kappa, ICC.

4. Validity

Validity = Does the instrument measure what it intends to measure?

Face validity, content validity, criterion validity (concurrent/predictive).

Construct validity: Convergent (high correlation with similar measures) + discriminant (low correlation with different measures).

5. Item and Scale Construction

Item formulation: Simple, short, unambiguous. No double negatives. No double-barreled items.

Response formats: Likert (1-5, 1-7), semantic differential, visual analog scale (VAS).

Effects: Social desirability, acquiescence, extreme response style.